

PROJECT mC

OPENBRAILLE

SEMESTER ARCHITECTURE

AND FUTURE RESEARCH REFERENCE

DOSSIER

Frozen Macro Six-Dot Build Plan, Validation Programme and Post-Semester VPD-C7 Research Path

Control item	Value
Course	PBCST504 - Microcontrollers Micro Project
Programme	B.Tech Computer Science and Engineering (Cyber Security)
Group	Group 8
Document status	Complete retained-reference and fallback implementation dossier
Current portfolio status	VibeGuard switch requested; OpenBraille remains the fully documented alternative
Frozen architecture	One macro six-dot cell; six SG90-class servos; eccentric cams; guided pins; ESP32 direct PWM
Expected semester budget	Approximately ₹2,984; conservative envelope approximately ₹4,909
Future mechanism	VPD-C7 research hypothesis - unbuilt, confidential and evidence-gated

Controlling decision

This dossier does not override the current request to adopt VibeGuard. It provides the complete OpenBraille plan if the teacher retains the zeroth-review concept, and preserves OpenBraille as a future accessibility/mechatronics research route.

Patent boundary

The macro six-servo semester prototype is conventional and is not represented as patent-ready. VPD-C7 is a separate post-semester hypothesis that must be redesigned, built, compared and professionally reviewed before any filing decision.

Prepared from the controlled Project mC authority set and the Master Evidence, Costing and Decision Basis. This document is an engineering reference plan, not a patent opinion, standards-compliance declaration or promise of grant.

Document control and approval use

Control item	Value
Architecture baseline	Frozen OpenBraille Phase 3C macro six-servo/cam architecture
Return-motion rule	Active servo-driven cam retraction unless a usable passive return is physically proved
Power baseline	External regulated 5 V servo supply; 5 V/3 A procurement target; measured droop controls acceptance
USB role	Serial command input and logs; not six-servo power
Semester boundary	No moving roller, encoder, RAMPS/Mega, continuous paper or electromagnetic substitution
Future boundary	VPD-C7 only on a separate post-MVP coupon; current never certifies pin height alone

Approval checkpoints required before an OpenBraille procurement freeze

1. Teacher confirmation that OpenBraille, rather than the requested VibeGuard alternative, remains the semester implementation.
2. Confirmation of the effective build window and dependable institutional FDM-printer access.
3. Approval of the one-dot-first mechanical validation plan before six servos and full-cell materials are committed.
4. Approval of the external 5 V power arrangement, pinch-point controls and tactile-participant protocol.
5. Approval of quantitative repeatability, cycle-life and pattern-test criteria before the locked validation run.

Executive reference decision

Complete alternative semester plan

OpenBraille is technically buildable within the ₹5,000 ceiling and offers a strong accessibility demonstration. Its execution risk is concentrated in FDM tolerances, cam/follower friction, active retraction, six-servo power behavior and trustworthy tactile-state validation. This dossier is therefore retained as the complete fallback/reference plan while VibeGuard remains the recommended semester choice.

The correct semester claim is narrow: one macro-scale six-dot cell accepts a character or dot pattern over USB/serial, commands six independent servo/cam channels, produces repeatable raised/lowered tactile states, and reports measured limitations. The project must not be presented as a standards-compliant commercial Braille display or as a patented actuator mechanism.

Success definition at a glance

Layer	Required semester result	Evidence
Mechanical	Six independently moving macro dots with active raise/retract and safe rounded pin tips	CAD/build IDs, caliper measurements, photographs, cycle logs
Electrical	Stable ESP32 control and external servo rail without brownouts or unsafe heating	Voltage/current observations, reset log, power checklist
Firmware	Correct character-to-six-dot mapping, per-dot calibration and USB commands/logs	Unit tests, configuration record, pattern test report
Validation	Repeatability, lifecycle, pattern-transition and approved tactile evidence	Locked test sheet, raw records, failed-run register
Scope honesty	Macro educational prototype; no physical-state claim from command timing alone	Explicit limitations in report and live demonstration
Future separation	VPD-C7 kept outside the semester architecture	Separate research folder and disclosure controls

1. Purpose, audience and scope

1.1 Purpose

This dossier converts the frozen OpenBraille Phase 3C architecture into a procurement-aware and execution-ready semester plan. It defines the hardware, mechanical build order, electrical integration, firmware, calibration, validation, team ownership, semester schedule and future-research boundary.

1.2 Audience

- Project coordinator and teacher: concept-retention decision, scope, budget, safety and review gates.
- Five-member student team: CAD, fabrication, firmware, assembly, testing, evidence and demonstration responsibilities.
- Department laboratories: FDM access, tools, metrology and safety support.
- Accessibility or tactile-test reviewers: honest macro-prototype scope and participant protocol.
- Institutional innovation/IPR reviewers: separation between the public semester cell and confidential VPD-C7 work.

1.3 Scope hierarchy

Scope tier	Included	Not included
Semester OpenBraille	One macro six-dot cell; six servos/cams/pins; direct PWM; active retraction; USB; measurements	Standards-compliant product, multi-cell line, production design, patent claim
Semester instrumentation	Calipers/camera, voltage/current checks and test scripts used as development evidence	Permanent endpoint contacts or current-only state inference
Post-semester VPD-C7	Separate coupon studying a datum that sets and directly evidences pin endpoints	Assumed performance, filing-ready claims or automatic inclusion in the semester cell

2. Engineering identity and problem definition

2.1 Engineering identity

OpenBraille is an assistive embedded-system experiment focused on lowering the cost of refreshable tactile output. Its semester purpose is not to compete with commercial displays, but to test whether a simple low-cost actuation channel can create one repeatable six-dot cell suitable for education and engineering study.

2.2 Semester engineering hypothesis

Hypothesis to test

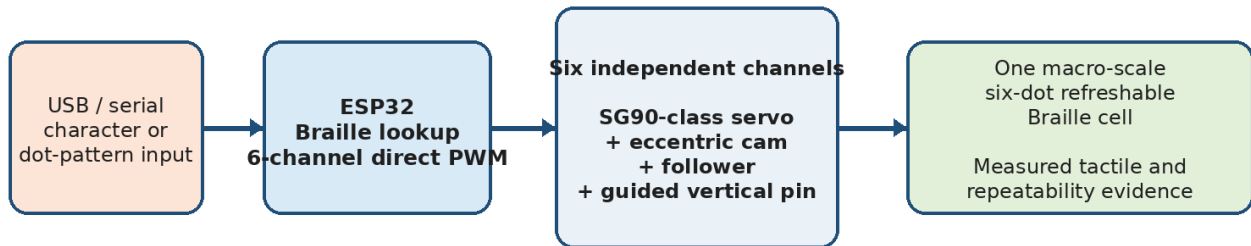
A macro-scaled servo-and-cam mechanism can produce repeatable raised and lowered tactile states for one six-dot Braille cell within the student budget and schedule, while accepting that the prototype is not commercial-scale or standards-compliant.

2.3 What the hypothesis does not claim

- It does not claim that servo + cam + Braille pin actuation is novel.
- It does not claim commercial Braille pitch, force, refresh speed, noise or lifetime.
- It does not assume a passive servo spring retracts the pin.
- It does not treat a PWM command, elapsed time or servo current as direct proof of tactile-pin height.
- It does not establish a patent or freedom to operate.

3. Frozen system architecture

Frozen OpenBraille Semester Architecture



Scope boundary: no roller, encoder, RAMPS/Mega embosser or electromagnetic substitution.

Semester result: macro six-dot prototype with active retraction and measured limits.
Educational evidence only - not standards compliance or a patent claim.

Figure 1. Frozen macro six-dot architecture and explicit semester scope boundary.

3.1 Subsystem boundaries

Subsystem	Inputs	Outputs	Owner acceptance
USB/serial interface	Character, pattern or test command	Parsed request, acknowledgement and log line	Commands are deterministic, versioned and unit-tested
ESP32 control	Pattern, per-dot calibration and fixed sequencing rules	Six PWM outputs and timestamped command records	No invalid GPIO use, unstable resets or hidden configuration
Six actuator channels	5 V servo power and PWM	Cam rotation and follower displacement	Each channel accepted on a one-dot fixture before full-cell assembly
Mechanical cell	Six channel motions	Six guided raised/lowered tactile pins	No binding, unsafe tips or adjacent-channel interference
Power distribution	External regulated 5 V and common ground	Stable servo rail and logic operation	No USB servo power, brownout or unsafe heating
Development measurement	Pin position, cycle count, rail behavior and operator observations	Evidence package and limitations	External measurement remains the truth source for physical position

3.2 Frozen architecture controls

- One macro-scale six-dot cell only.
- Six SG90-class servos, one eccentric cam and one guided vertical pin per dot.
- ESP32 local direct-PWM control; no intermediate motor driver is required for the servo signal.
- FDM frame/guides/cams and external regulated servo power.
- Active servo-driven retraction unless a procured mechanism proves a reliable passive return under load.
- No moving roller, encoder wheel, RAMPS/Mega embosser, continuous-paper system or electromagnetic substitution.

4. Detailed hardware architecture and BOM

4.1 Required purchased and fabricated items

Item	Qty	Exact semester role	Expected	Conservative	Acceptance/ procurement rule
ESP32 development board	1	Six PWM channels, USB/serial commands and logging	₹400	₹899	Verified WROOM-32 DevKit baseline; ESP32-S3 authorized fallback. Record pinout and regulator.
SG90-class servo	6	One independent eccentric-cam actuator per dot	₹120 each / ₹720	₹142 each / ₹852	Buy one consistent batch where possible; inspect every unit; no passive-return assumption.
5 V / 3 A regulated adapter	1	External servo supply with test headroom	₹289	₹408	Connector and regulation verified; adequacy controlled by measured droop and stall tests.
1,000 μ F electrolytic capacitor	1-2	Servo-rail transient buffering	₹25	₹50	Install near distribution; voltage rating above 5 V; not a substitute for an adequate supply.
FDM frame, guides, cams and followers	1 cell	Macro mechanical structure	₹450	₹900	Expected case assumes institutional printer access; print a single-dot coupon first.
Stainless pins/rod with rounded tips	6 + spare	Guided tactile dots	₹250	₹400	Local quote; polished/rounded, safe for repeated touch, dimensions recorded.
Fasteners, servo mounts and cable management	1 lot	Alignment, assembly, strain relief and guards	₹300	₹500	Final quantity follows CAD and physical servo dimensions.
Perfboard, wires, headers and connectors	1 lot	Power distribution and stable PWM wiring	₹200	₹350	Servo current never routed through ESP32 3.3 V rail or thin USB wiring.
USB data cable	1	Serial input and logs	₹100	₹150	Reusable cable acceptable after connection reliability check.

Budget layer	Low	Expected	Conservative
Required hardware/material subtotal	₹1,866	₹2,734	₹4,509
Delivery/local variability/replacement allowance	₹100	₹250	₹400
Semester planning total	₹1,966	₹2,984	₹4,909

Budget decision

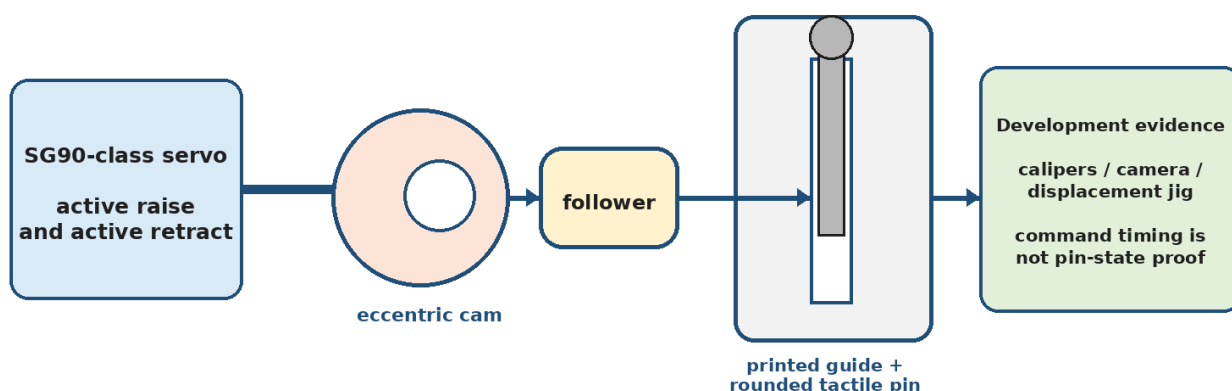
Use approximately ₹3,000 as the expected teacher-facing budget. The conservative case remains below ₹5,000 only when institutional printing is available and full-cell reprints are limited. One or two spare servos are useful but should be approved separately.

4.2 Reusable or institutional equipment

Equipment	Use	Required status
Laptop with Arduino IDE and Python	Firmware, serial commands, analysis and reporting	Assumed available
FDM printer and slicing software	Frame, guides, cams, followers and coupons	Material dependency; access must be confirmed
Digital calipers	Pin travel, spacing, clearances, repeatability and wear	Required
Multimeter	Power, continuity and servo-rail checks	Required
Bench supply with current limit	One-servo bring-up and fault-safe testing	Strongly recommended / borrowed
Oscilloscope or logic analyzer	PWM and rail-transient checks	Useful but not essential
Camera / fixed measurement jig	External visual ground truth and configuration record	Required for credible evidence
Basic hand tools and soldering station	Assembly, finishing and final wiring	Institutional/reusable
Tactile participant access and approved protocol	Raised/lowered and pattern distinguishability	Not confirmed; teacher/ethics decision

5. Mechanical architecture and one-dot-first build

One-Dot Mechanical Channel: What Must Physically Work



Control rule: use active cam retraction unless a procured mechanism proves a reliable passive return under load.

Figure 2. One-dot channel and the separation between command, physical motion and development ground truth.

5.1 Mechanical design rules

1. Start with one independently testable servo + cam + follower + pin coupon. Do not print the full cell before the coupon moves freely.
2. Use macro spacing and travel as a documented educational design choice. A starting target near 300% of standard sign-Braille geometry may guide CAD, but the prototype must not be labelled standards-compliant.
3. Use rounded, smooth tactile tips and guard all pinch points around cams and servo horns.
4. Set pin-guide clearance through printed tolerance coupons and measured finishing; nominal CAD dimensions do not prove physical clearance.
5. Use active cam-driven raise and retract positions. Do not rely on an unverified servo spring or gravity return.
6. Record each printed part revision, orientation, material, measured dimensions, sanding/finishing and assembly operator.
7. Treat each channel as replaceable for test purposes, but do not present modular replacement as a novel patent feature.

5.2 Provisional geometry and acceptance parameters

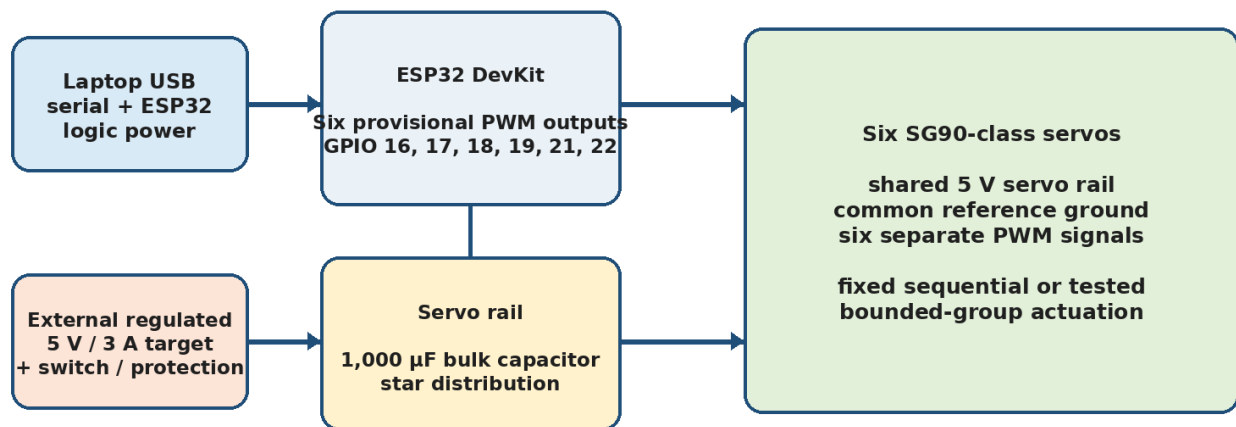
Parameter	Starting design basis	Acceptance rule
Cell scale	Macro single cell, approximately 3x commercial sign-Braille pitch as an engineering starting point	Final CAD and measured cell clearly labelled non-standard
Raised travel	Approximately 1.5 mm starting target	Teacher-approved tactile and repeatability test controls final value
Pin material	Approximately 2 mm stainless rod with rounded tip	No sharp edge, binding, corrosion or unsafe surface
Cam throw	Set through one-dot CAD/coupon to create required travel with margin	No stall, follower loss or adjacent interference
Guide clearance	Derived from printed tolerance coupon	Free motion under the selected load with repeatable alignment
Actuation sequence	Fixed sequential or bounded group of changed dots	No brownout, unsafe peak or excessive refresh delay

5.3 Mechanical stop conditions

- Pin binds, tilts, scrapes or fails to retract fully.
- Cam/follower loses contact, jumps, cracks or creates an exposed pinch hazard.
- Servo stalls or heats during the normal pattern sequence.
- Pin height varies beyond the approved repeatability band.
- A rounded tactile tip becomes loose or unsafe.
- Full-cell assembly changes the motion that passed on the single-dot coupon.

6. Electrical integration and provisional pin map

Power, USB and Six-Channel Control Boundary



Never power the six servos from laptop USB or the ESP32 3.3 V regulator.

Figure 3. Recommended power and control boundary. Final wiring must be verified against the purchased board and supply.

6.1 Power domains

Domain	Source	Loads	Rule
Logic / USB	Laptop USB to ESP32 DevKit	ESP32 and serial interface	Do not back-feed a second 5 V source into an unverified board while USB is connected
Servo rail	External regulated 5 V / 3 A target	Six SG90-class servos	Separate branch and conductors sized for servo current; bulk capacitor near distribution

Domain	Source	Loads	Rule
Signal reference	Common ground	ESP32 PWM and servo signal reference	Use star/controlled grounding; inspect for resets and noise
Measurement	Multimeter / optional scope	Servo rail and logic rail	Measurements must not create unsafe shorts or loose probes near moving cams

6.2 Provisional ESP32-WROOM-32 pin assignment

Function	Provisional GPIO	Notes
Dot 1 PWM	GPIO 16	Output-capable; verify board label
Dot 2 PWM	GPIO 17	Output-capable; verify board label
Dot 3 PWM	GPIO 18	Output-capable; avoid conflict with other planned peripherals
Dot 4 PWM	GPIO 19	Output-capable
Dot 5 PWM	GPIO 21	Output-capable
Dot 6 PWM	GPIO 22	Output-capable
USB serial	Onboard USB-UART	Primary command/log interface
Optional status LED	GPIO 23 or onboard LED where available	Not part of the six-dot mechanism; verify polarity/board mapping

Implementation note

This is a Phase 4 wiring proposal, not a source-file mandate. Revalidate every pin against the exact purchased DevKit and do not use boot-strapping or input-only pins without understanding their behavior.

6.3 Wiring acceptance checklist

- ESP32 boots and USB serial works with servo rail disconnected.
- External adapter polarity and no-load/loaded voltage are verified.
- Servo rail and logic share only the intended reference ground.
- Each servo is tested alone before joining the distribution rail.
- Bulk capacitor polarity and placement are correct.
- No servo current passes through the ESP32 3.3 V regulator or laptop USB cable.
- Combined pattern transitions produce no reset, cable heating or connector instability.

7. Firmware and software architecture

7.1 Firmware module breakdown

Module	Responsibility	Required outputs/tests
Configuration manager	Board variant, pin map, cell/build ID, PWM calibration and sequence policy	Startup banner and configuration record
USB command parser	Accept character, six-bit pattern, clear, test and calibration commands	Malformed-command tests and deterministic responses
Braille mapping table	Map supported characters to six-dot patterns	Unit test independent of hardware
PWM/channel manager	Apply per-dot up/down pulse widths and fixed sequencing	Logic-analyzer/servo acceptance and no illegal overlap
Calibration manager	Store per-dot command values established with external measurement	Versioned calibration table and reset procedure

Module	Responsibility	Required outputs/tests
Pattern transition engine	Move only changed dots, apply settle interval and update commanded state	Transition log and pattern suite
Fault/log manager	Record resets, invalid commands, manual faults and channel disable state	Human-readable USB log; no false physical-state claim
Test mode	Cycle one channel or selected patterns under operator control	Repeatability/lifecycle scripts

7.2 Repository and configuration structure

Folder/file	Purpose
firmware/	ESP32 source, board configuration and release tags
firmware/config/	Pin map, per-dot calibration and build identifiers
firmware/tests/	Character mapping and command-parser unit tests
cad/	Versioned frame, guide, cam, follower and coupon designs
fabrication/	Slicer settings, material, orientation and post-processing record
protocols/	Electrical, mechanical, cycle, tactile and demo procedures
data/	Measurements, cycle logs, power observations and fault records
evidence/	Photographs, videos, signed checklists and final locked results

7.3 Command and state semantics

The firmware should distinguish commanded state from verified physical state. Without a permanent pin sensor, the runtime may record that a PWM command and settle interval completed, but it must not label that event UPPER_CONFIRMED or LOWER_CONFIRMED. Physical confirmation belongs to external development measurement and the approved tactile protocol.

State	Meaning	Allowed evidence	Exit/exception
STARTUP	Board and configuration initialization	Firmware and board checks	FAULT on invalid configuration
READY	No active transition; accepts command	Commanded pattern record	RAISING / LOWERING / TEST
RAISING	One or more selected channels commanded toward up position	PWM command and timestamp	SETTLE_WAIT after command interval
LOWERING	Selected channels commanded toward down position	PWM command and timestamp	SETTLE_WAIT after command interval
SETTLE_WAIT	Fixed mechanical settling delay	Timer only	COMMAND_FINISHED; not physical confirmation
COMMAND_FINISHED	Requested command sequence ended	Command log	External tests decide physical correctness
FAULT/ISOLATED	Manual/operator or electrical/mechanical fault recorded	Fault input, reset, test observation	Operator-cleared after inspection

8. Calibration, sequencing and control logic

8.1 Per-dot calibration process

1. Assign a permanent channel and servo identity.
2. Use external calipers/camera or a fixed displacement jig to measure the lowered and raised pin positions.
3. Sweep the up command in small controlled increments to find the minimum repeatable raised position without stall or excessive load.
4. Repeat for active retraction and verify the pin returns to the approved lowered band.

5. Repeat across multiple cycles and store the selected command values with hardware/build IDs.
6. Invalidate or recheck calibration after servo replacement, cam/guide reprint or major reassembly.

8.2 Fixed sequencing policy

The semester system should use a simple fixed sequence: identify changed dots, lower dots that must clear, then raise dots that must set, with a bounded number of simultaneously moving channels chosen only after measured power testing. Dynamic current-aware sequencing belongs to future research and must not be backfilled into the frozen MVP.

8.3 Pattern transition rules

Rule	Purpose
Move only changed dots	Reduces unnecessary cycles, noise and power
Active retract before conflicting raise	Prevents ambiguous intermediate patterns
Fixed settle time per tested channel group	Makes behavior reproducible and loggable
Reject malformed/unsupported command	Protects mapping and state integrity
Manual stop on stall, heating or binding	No automatic recovery without evidence
Calibration version stored with every test	Prevents results from being separated from command values

8.4 Physical-truth boundary

Do not overclaim state

A servo angle, PWM pulse, elapsed time, current rise or visible cam movement does not automatically prove acceptable tactile-pin height. Semester physical truth comes from external measurement and an approved tactile evaluation. Current sensing may be used for power/safety observation, not endpoint certification.

9. Data and evidence management

9.1 Required build and session metadata

- Project, firmware, CAD, slicer and protocol version.
- Servo identity, seller/batch, channel assignment and replacement history.
- Cam, follower, guide and pin revision; material, print orientation and finishing.
- Per-dot up/down command values and calibration date.
- Supply identity, measured voltage, actuation grouping and reset/brownout observations.
- Pattern sequence, cycle count, operator, temperature/contamination/load note where relevant.
- Measured raised/lowered positions, failed transitions, binding, wear and manual interventions.
- Photographs/videos tied to the exact configuration ID.

9.2 Evidence hierarchy

Evidence type	What it supports	What it does not support
USB command log	Requested patterns, timestamps and firmware behavior	Actual tactile height
Calipers/camera/displacement jig	Development measurement of pin positions	Long-term tactile usability by itself
Power measurements	Supply adequacy, peaks, droop and resets	Endpoint correctness
Cycle log	Repeatability and mechanical failure history	Commercial lifetime
Tactile protocol	Participant distinguishability under the approved setup	Universal user performance or standards compliance
Photographs/video	Configuration and visible operation	Precise dimensions without measurement

9.3 Final evidence package

Evidence file	Minimum content
Configuration record	Board, pins, six servo IDs, CAD/slicer versions, power source and calibration table
Mechanical acceptance	One-dot and six-dot dimensions, travel, free-motion and safety checklist
Electrical acceptance	Supply voltage, resets, wiring, capacitor and combined-pattern observations
Pattern test	Supported character/pattern suite and exact pass/fail outputs
Repeatability/lifecycle	Per-channel measurements, cycle counts and failures
Tactile protocol report	Approved method, participants/ethics status, randomization, results and limitations
Failure register	Binding, wear, servo replacement, reprints, rejected runs and corrective action
Final demo package	Frozen firmware/CAD, demo script, backup evidence and limitations

10. Validation and acceptance programme

10.1 Validation stages

Stage	Question	Minimum evidence	Stop/fallback
OB-V0 Component acceptance	Are the ESP32, first servo, supply and tools usable?	Board/USB check, servo sweep, adapter polarity/voltage	Replace failed component before CAD freeze
OB-V1 One-dot mechanics	Can one printed channel raise and actively retract without binding?	Measured motion, photographs and repeated cycles	Redesign guide/cam before ordering full batch
OB-V2 Calibration	Can per-dot command windows produce repeatable positions?	External measurement across cycles	Change geometry or actuator; do not tune blindly
OB-V3 Six-dot integration	Do six channels coexist without interference and brownout?	Pattern commands, power observation and mechanical inspection	Use fixed sequencing, stronger supply or rebuild distribution
OB-V4 Pattern correctness	Does character mapping produce intended six-dot states?	Unit tests and physical pattern checks	Fix mapping/configuration before user testing
OB-V5 Repeatability/lifecycle	Does the mechanism retain position and motion over approved cycles?	100-cycle precision set and at least 1,000 total actuations	Record failure; redesign or limit claim
OB-V6 Tactile evaluation	Are raised/lowered dots and selected patterns distinguishable?	Teacher-approved blinded protocol	Adjust geometry or report MDS not achieved
OB-V7 Locked demonstration	Can the frozen build reproduce the result safely?	Signed checklist, frozen configuration and final evidence	Use approved backup evidence; no late redesign

10.2 Proposed semester acceptance targets for teacher approval

Status of numbers The project authority contains some architecture-level repeatability and lifecycle targets, but the tactile participant threshold is not frozen. The following set should be preregistered and approved before the locked run.		
Metric	Proposed target	Why it matters
Channel function	All six channels independently raise and actively retract throughout the final pattern sequence	Minimum six-dot cell functionality
Repeatability	≤ 0.1 mm position variation over 100 measured cycles under the approved setup	Source-derived architecture target; measures mechanical consistency
Lifecycle screen	At least 1,000 actuations without mechanical failure that invalidates the demo	Early engineering screen, not product lifetime

Pattern mapping	100% correct output for the frozen supported character/test set	Separates firmware mapping from mechanical failure
Power integrity	No brownout/reset, connector heating or unsafe voltage collapse in the locked sequence	Electrical safety and reliability
Tactile distinguishability	Teacher-approved blinded threshold and trial count fixed before testing	Avoids inventing or post-hoc selecting a success rate
Failure honesty	Every incomplete/bound transition is logged; no command-complete event relabelled as physically verified	Evidence integrity
Budget	Final semester spend $\leq$ ₹5,000, with preferred target near ₹3,000	Course requirement

10.3 Required performance reporting

- Per-dot raised and lowered position distributions, not only average travel.
- Pattern-level pass/fail counts and the exact supported character set.
- Power/reset observations for one, multiple and worst planned transitions.
- Cycle count, wear inspection and parts replaced.
- Tactile results with protocol, participant context and limitations.
- Negative evidence, binding, failed prints and calibration changes.
- Explicit statement that the macro prototype does not establish commercial-scale feasibility.

11. Twelve-week semester roadmap

Week	Primary work	Owner(s)	Measurable output	Exit criterion	Fallback
1	Teacher decision, scope freeze, FDM/tool audit and first procurement	Nihad + team; Amith fabrication audit	Approved concept status; one-servo order; printer/tool booking; risk register	No full-cell purchase until one-dot gate and teacher direction are confirmed	Use available verified ESP32/servo variants without changing architecture
2	ESP32/USB bring-up, one bare servo and power acceptance	Sreehari + Amith	Commanded servo sweep; adapter/polarity record; provisional pin map	Stable board/serial and safe one-servo motion	Replace failed board/servo/supply before CAD integration
3	Print tolerance coupons and one-dot frame/cam/follower	Amith + Nihad	Measured coupon set; one-dot assembly; CAD/build ID	Pin moves freely and cam produces controllable travel	Adjust guide/cam dimensions; laser-cut guide fallback only if approved
4	Active raise/retract calibration and single-dot repeatability	Amith + Sreehari; Sreeprada tests	Per-dot calibration procedure; 100-cycle single-dot dataset	Stable raised/lowered bands without binding or unsafe stall	Change geometry/servo before multiplying channels
5	Procure/accept remaining servos and fabricate six-dot parts	Amith + Sreeprada inventory	Six accepted servo IDs; completed printed parts; assembly checklist	Each channel passes independent bench test	Use spares or pause channels; do not hide batch variation
6	Six-dot mechanical and electrical integration	Nihad + Amith + Sreehari	Assembled cell; external rail; six PWM channels; command logs	All six move independently; no brownout/interference	Fixed sequential actuation, supply/distribution revision or rebuild
7	Braille mapping, pattern transitions and per-dot calibration	Sreehari + Sreeprada	Supported character set; mapping unit tests; calibration table	Frozen test patterns produce correct commanded states	Reduce demo set while keeping mapping evidence honest
8	Power, safety, repeatability and failure-mode testing	Amith + Nihad; Sreeprada runs	Worst planned transition record; fault/reprint log; safety review	No unsafe heating, reset, pinch or uncontrolled motion	Limit actuation group, strengthen guard/supply or redesign
9	Lifecycle screen and tactile-protocol approval	Nihad + Archa; teacher/participant coordination	Cycle progress; approved blinded protocol or documented access limit	Mechanism survives and protocol is approved before use	Report tactile validation unavailable rather than improvising
10	Locked pattern/repeatability/tactile validation	Full team; test led by Sreeprada	Frozen configuration and signed result sheets	Approved targets met without post-result calibration changes	Record failure; use honest partial-MDS conclusion
11	Final report, source records and live-demo rehearsal	Archa + Nihad; technical review by Sreehari/Amith	Final BOM, CAD/firmware versions, results, limitations and demo script	Every technical claim traces to evidence	Remove unsupported claims and use backup video only if approved

Week	Primary work	Owner(s)	Measurable output	Exit criterion	Fallback
12	Contingency, repair-only window and final delivery	Full team	Stable final build and archived evidence package	No late VPD-C7 or architecture expansion	Submit the strongest verified semester result

Schedule rule

The minimum demonstrable cell should work by the end of Week 8. Weeks 9-12 are for lifecycle, tactile evidence, recovery and presentation. VPD-C7 hardware must not enter the semester critical path.

12. Team roles, actual members and accountability

Role status

The following map mirrors the established five-member structure. It is a proposed assignment for team approval; it deliberately gives mechanical work substantial ownership because that is OpenBraille's critical risk.

Member	Proposed primary role	Specific OpenBraille ownership	Review/backup obligation
Nihad P C - JEC24CC044	Technical integration lead / project manager	Architecture control, schedule, interface decisions, validation protocol, final integration and evidence integrity	Backs up firmware and hardware; approves configuration freezes
Sreehari K - JEC24CC055	Firmware and embedded-control lead	USB command parser, Braille mapping, PWM/channel manager, calibration storage, test scripts and logs	Nihad reviews releases; Sreeprada executes scripted tests
Amith Krishna Das - JEC24CC016	Mechanical, hardware and power lead	CAD/FDM, cam/follower/guide design, servo acceptance, power rail, assembly and safety inspection	Nihad reviews interfaces/safety; Sreehari checks control requirements
Sreeprada K S - JEC24CC056	Learner, test and inventory assistant	Inventory, servo/build labels, cycle scripts, measurements, photos, test registry and repeated independent trials	Works from signed procedures; learns USB operation and calibration workflow
Archa Pramod - JEC24CC022	Documentation, teacher communication and demonstration lead	Weekly records, teacher-safe explanations, tactile-test coordination, live-demo script, final presentation and source register	Technical statements reviewed by Nihad, Sreehari and Amith

12.1 Responsibility rules

- Every critical subsystem has one primary owner, one technical reviewer and one written checklist.
- No one changes CAD, PWM calibration, power wiring or test labels without a dated configuration update.
- The learner/test role is evidence-critical and must run independent repetitions, not only observe.
- The documentation lead does not publish VPD-C7 details without technical and IPR review.
- Nihad acts as integration failsafe, but CAD, firmware and validation must remain independently owned and reviewable.

13. Procurement, acceptance and configuration control

13.1 Procurement sequence

1. Confirm teacher decision and institutional FDM access.
2. Buy one verified ESP32 board, one SG90-class servo, the external supply and basic mechanical material first.
3. Pass the one-dot mechanical/active-retraction gate before buying or committing the remaining servo batch and full-cell prints.
4. Buy the remaining servos from one consistent seller/batch where practical and record every identity.
5. Fabricate the full frame only after the physical servo and pin dimensions are measured.
6. Purchase spares only when acceptance or delivery risk justifies them within the conservative envelope.

13.2 Component acceptance record

Component	Acceptance checks	Reject if
ESP32	Reliable boot/USB, exact board identity and usable output GPIOs	Unknown pinout, repeated disconnect, overheating or unstable regulator
SG90 servo	Smooth sweep, no damaged gears, stable horn/shaft and repeatable response	Dead zone, severe jitter, gear damage, overheating or inconsistent return command
5 V adapter	Correct polarity, stable loaded output, adequate connector and no overheating	Unsafe voltage, instability, excessive droop or poor connector
FDM coupon	Measured clearance, free pin motion, no cracks or warped datum surfaces	Binding, unsafe roughness, large warp or non-repeatable fit
Cam/follower	Smooth engagement and required travel without losing contact	Stiction, follower jump, crack or unsafe pinch exposure
Pin/tip	Straight, smooth, rounded and securely retained	Sharp edge, looseness, corrosion or unsafe touch surface

13.3 Configuration ID format

Recommended format: OB-[cell/build]-[dot/servo]-[CAD]-[firmware]-[protocol]-[session]. Example: OB-C01-D3-SV05-CAD0.4-FW0.6-P03-R012. The exact format may change, but every measurement, image and result must identify the physical and software configuration.

14. Risk register and mitigation

Risk	Likelihood / impact	Early indicator	Mitigation	Decision consequence
FDM guide/cam tolerance and friction	High / High	Pin does not slide freely; variable travel; follower drag	One-dot coupons, measured clearance, material/print iteration, limited finishing	Do not proceed to six-dot assembly until passed
No usable passive return	High / Medium	Pin remains raised or servo does not return under load	Use active commanded retraction; remove passive-return language	Architecture remains valid with corrected control description
Servo batch variation/backlash	Medium / High	Different travel for common PWM; jitter or gear play	Consistent batch, per-dot calibration, spare/replace failed unit	May increase calibration and evidence burden
Six-servo brownout or rail droop	Medium / High	ESP32 reset, adapter sag, connector heating	Separate 5 V rail, bulk capacitor, fixed sequencing, measured load	Upgrade supply/distribution or reduce simultaneous motion
Pin tip unsafe or tactility poor	Medium / High	Sharp/loose tip or participant cannot distinguish states	Rounded tips, geometry iteration, approved tactile protocol	MDS may fail even when motion is visible
Cam/follower wear	Medium / Medium	Position drift, dust, roughness or increased noise	Material choice, inspection, lifecycle screen, replaceable coupon	Limit claims and redesign if systematic
CAD/print schedule slippage	High / Medium	Multiple full reprints or printer unavailable	Single-dot gate, modular prints, early lab booking, controlled fallback	May force reduced demo or switch to VibeGuard
Firmware mapping/calibration error	Low / Medium	Wrong pattern despite correct mechanics	Independent unit tests and configuration versioning	Fix before tactile validation
Tactile participant/protocol unavailable	Medium / Medium	No approved access or trial design	Secure approval early; document visual/mechanical evidence separately	Do not claim tactile success without the test
Macro prototype overstated as product	Medium / High	Slides imply standard compliance or commercial readiness	Teacher-safe wording, explicit dimensions/limits and review	Remove unsupported claim before presentation
Future patent concept leaks into MVP	Medium / Medium	VPD-C7 contact/current details added late	Separate folders, approval gate and confidentiality rule	Exclude from semester build and results

15. Required semester deliverables and live demonstration

15.1 Deliverables

- Working macro one-cell, six-dot servo/cam OpenBraille prototype if the concept is retained.
- Final BOM, procurement record, power diagram and provisional pin map.
- Versioned CAD, slicer settings, firmware and Braille mapping tests.
- One-dot and six-dot mechanical acceptance records.
- Per-dot calibration table and pattern-transition test suite.
- Repeatability, lifecycle, power and tactile-protocol results with limitations.
- Failure/reprint/replacement register and contributor chronology.
- Teacher-ready presentation and live-demo procedure.
- Separate confidential VPD-C7 research folder with no semester performance backfill.

15.2 Recommended five-minute live demonstration

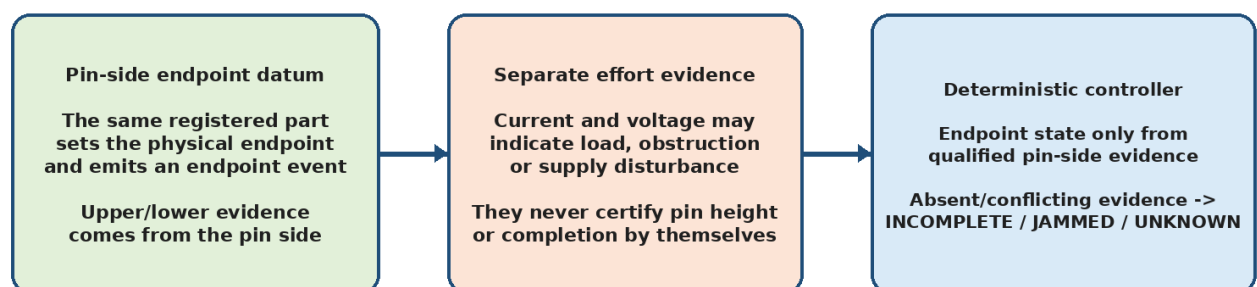
1. Show the powered-off macro cell, six servos, rounded pins, guard/covering and separate servo power rail.
2. Connect USB, display the configuration banner and run a commanded one-dot test while explaining that command timing is not physical confirmation.
3. Display a small frozen sequence of characters or six-bit patterns and clear/refresh them.
4. Show per-dot calibration and one repeatability result table or measurement image.
5. Run the approved safe pattern transition while showing no reset/brownout and the USB log.
6. State the limitations: macro scale, educational prototype, active retraction, no standard-compliance or patent claim.

15.3 Demo failure fallback

If classroom operation is unsafe, unreliable or too noisy, use an approved guarded video of the locked physical setup plus a live USB/firmware demonstration and the signed measurement evidence. The fallback must not substitute a different actuator or conceal that the physical build failed the planned gate.

16. Post-semester VPD-C7 research hypothesis

Post-Semester VPD-C7 Research Hypothesis



Required comparison: passive stop + open loop | current only | separate direct sensor | full VPD-C7

Status: unbuilt and evidence-gated; it must show a distinct physical advantage to survive obviousness attacks.

Figure 4. VPD-C7 research concept. It is not implemented or validated in the semester cell.

16.1 Technical question

Can a pin-side mechanical datum that sets the tactile endpoint also provide direct endpoint evidence with less registration error and fewer silent wrong states than a passive stop plus a separately mounted sensor, while keeping actuator current limited to independent effort/supply information?

16.2 Candidate mechanism in non-claim language

1. Redesign one post-MVP guide/pin coupon with mechanically registered upper and lower endpoint features.

2. Arrange the endpoint feature so reaching it can produce a low-energy pin-side event without carrying servo power through the contact.
3. Use external displacement and force measurement as the development oracle; do not let the candidate sensor label itself as truth.
4. Use a separate current/voltage channel only to characterize effort, obstruction-like behavior or supply disturbance.
5. Declare an endpoint state only from qualified pin-side evidence; absent or conflicting evidence becomes incomplete, jammed or Unknown.
6. Permit recovery/isolation only after the physical and safety behavior is experimentally qualified.

16.3 Why this is only a hypothesis

- A mechanical stop that also acts as a switch/contact is familiar engineering and may be obvious.
- The added contact may increase friction, force, bounce, contamination sensitivity or wear.
- Practical Braille pitch may not accommodate the contact and wiring.
- A separately mounted optical/magnetic sensor may perform just as well or better.
- Current may add no independent diagnostic value beyond command, time and voltage.
- The official Indian close-prior-art file and worldwide claims remain unresolved.
- No experiment currently shows a cooperative or non-additive physical effect.

17. Six-to-twelve-month future research roadmap

Month	Stage	Work	Pass evidence	Failure action
0-2	Semester foundation and search boundary	Complete frozen macro MVP; preserve mechanical/power records; obtain official close-prior-art documents; build direct displacement oracle	Stable one-dot baseline and a defined professional search problem	Continue semester/publication only if close art anticipates the structural core
2-4	Current observability and direct sensor baseline	Compare calibrated open loop, current-only observation and a separate direct pin sensor on one-dot fixtures	Held-out evidence showing what current can and cannot reveal	Remove current from the putative invention if it adds no independent value
4-6	Dual-purpose datum coupon	Build passive-stop, separate-sensor and same-datum-contact coupons; measure force, bounce, registration and false endpoints	Distinct physical advantage fixed before confirmatory data	End VPD-C7 thesis if separate sensor performs equivalently
6-8	Practical-scale and contamination tests	Develop smaller coupons; test wiring, force, dust/fiber, cleaning and assembly yield	Feasible packing and stable endpoint evidence	Hold Track D if only macro geometry works
8-10	Replacement, reassembly and lifecycle	Blind servo/cam/pin/datum swaps; multiple operators; extended cycling and inspection	Transfer across builds with bounded drift and no systematic false endpoint	Redesign and restart after systematic wear/failure
10-11	Locked held-out evidence	Run the preregistered 3,000-transition silent-error test on unseen units/builds/conditions	Zero silently accepted wrong states plus practical Unknown/false isolation rates	Kill the route after a silent wrong state unless a genuinely new mechanism restarts the gate
11-12	Professional decision gate	Refresh worldwide search; claim/FTO chart; inventorship/disclosure review; management file/no-file decision	Useful narrow scope supported by reproducible physical evidence	Choose confidential know-how, defensive/academic publication or stop

17.1 Research baselines and ablations

Type	Required comparison
Strong conventional baselines	Fixed common PWM; calibrated per-dot open loop; passive datum only; direct separate sensor; fixed sequential refresh

Type	Required comparison
Current tests	Current without endpoint contact; endpoint contact without current; current removed from full candidate
State logic	Unknown retained versus coerced binary decision; retry removed; isolation removed
Service robustness	Servo/pin/cam/datum replacement and independent reassembly
Scale and environment	Macro versus practical-scale coupons; clean versus contamination/wear conditions

17.2 Patent-development kill conditions

- A closer enabling reference or official Indian file discloses the defining combination.
- The same-datum contact has no measurable advantage over a passive stop plus separate sensor.
- The contact adds unacceptable force, friction, bounce, wear or debris sensitivity.
- Practical-scale packaging fails.
- Performance collapses after servo replacement, reprint, remount or reassembly.
- Current adds no independent diagnostic value or is required to certify endpoint completion.
- The locked 3,000-transition test contains a silently accepted wrong state.
- Useful scope collapses to exact hobby parts, macro dimensions or a trivial retry count.
- Institutional/professional review does not justify filing cost or disclosure risk.

18. Confidentiality, inventorship and disclosure control

18.1 Public-safe semester material

- Macro six-dot architecture, six SG90-class servos, eccentric cams, guided pins, ESP32 direct PWM and external servo power.
- Semester budget, team roles, CAD/fabrication learning, repeatability/lifecycle and approved tactile results.
- Honest limitations: macro educational prototype, active retraction, no current-only state verification and no patent claim.

18.2 Keep private until IPR review

- Exact VPD-C7 contact/datum geometry, materials, dimensions, circuits and thresholds.
- Unpublished comparative results against passive stops, separate sensors and current-only inference.
- Replacement/reassembly, wear, contamination and held-out silent-error datasets.
- Claim-to-experiment maps, filing strategies and contributor/inventorship analysis.

18.3 Inventorship/contribution record

Maintain dated records of who conceived each later mechanical/electrical interaction, who designed each experiment, who created CAD/firmware, and when the mechanism was first built and tested. AI/model assistance is not inventorship and must not replace human conception records.

18.4 Disclosure rule

Do not rely on an assumed grace period

Before publishing code, CAD, datasets, posters, videos, papers, competition entries or VPD-C7 details, obtain institutional IPR review. A patent filing or defensive publication does not create freedom to operate.

19. Decision gates and escalation rules

Gate	When	Pass	Fail response
OB-G0 Teacher concept decision	Before procurement	OpenBraille retained with scope/budget approved	Proceed with VibeGuard or revise the project request

Gate	When	Pass	Fail response
OB-G1 Facility/procurement	Week 1	Printer/tools and first components available	Do not commit full-cell schedule; use VibeGuard fallback if needed
OB-G2 One-dot mechanics	Weeks 3-4	Active raise/retract and repeatability pass	Redesign coupon; stop six-dot expansion
OB-G3 Six-servo power	Weeks 6-8	No unsafe droop, reset, heating or stall in planned sequence	Upgrade supply/distribution or reduce groups
OB-G4 Six-dot MDS	Week 8	Correct pattern transitions and safe operation	Report partial prototype or switch according to teacher direction
OB-G5 Locked validation	Week 10	Approved repeatability/lifecycle/tactile gates pass	No patent or performance overclaim; document failure
OB-G6 VPD-C7 research entry	Post-semester	Separate design, confidentiality and search approval	Keep OpenBraille as semester/publication record only
OB-G7 Counsel-led filing evaluation	Months 11-12	Unexpected physical effect, enabling evidence and useful scope survive review	Do not file; choose publication/know-how/stop

20. Open questions and teacher decisions

ID	Question / decision	Owner	Required by
Q1	Should OpenBraille remain the semester project or should the requested VibeGuard switch be approved?	Project coordinator	Before procurement
Q2	Is the working build window approximately eight weeks or the full twelve-week plan?	Teacher/project coordinator	Week 1
Q3	Is institutional FDM access dependable, and what material/print limits apply?	Amith + lab	Week 1
Q4	Which exact ESP32 and SG90 seller/batch are approved?	Nihad/Sreehari/Amith	Before order
Q5	What quantitative tactile-identification protocol and participant access are approved?	Teacher + Archa/Nihad	Before tactile test
Q6	What is the approved raised/lowered position band for the macro cell?	Technical team + teacher	Before locked validation
Q7	Is the 5 V / 3 A target supply adequate under the actual six-servo sequence?	Amith + Nihad	Before full integration
Q8	What content may be disclosed publicly before institutional IPR review?	IPR cell / project owner	Before publication or competition
Q9	Should VPD-C7 remain only a confidential future option after the semester?	Team + institution	After MVP evidence

21. Teacher approval and conditions

Decision item	Teacher response
Semester project	Retain OpenBraille / Approve VibeGuard switch / Revise and resubmit
Frozen OpenBraille scope	Approved / Approved with conditions / Not approved

Decision item	Teacher response
Expected budget near ₹3,000	Approved / Revise
External 5 V servo supply and safety controls	Approved / Lab review required
One-dot-first procurement gate	Approved / Revise
Tactile participant protocol	Approved / Separate ethics/access review required
Future VPD-C7 treatment	Confidential post-semester only / Not approved / Further review
Teacher conditions	
Signature and date	

22. Source basis

This dossier was prepared from the controlled Project mC master package. The main evidence layers are:

Evidence layer	Controlled source
Project authority	Engineering Design Review; Project mC Decision Register v1.2; approved Phase 3C Portfolio Closure Memo
Common evidence	Concept Evidence Matrix; Uncertainty and Test Register
OpenBraille technical authority	Final OpenBraille Architecture Report; Memory; SOP
Comparative decision context	Two final cross-concept adjudications - treated as context, not votes
Future research boundary	OpenBraille Pre-Build Future Patent-Case Simulation
Costing and teacher context	Project mC Teacher Documents - Master Evidence, Costing and Decision Basis
Previous presentation	Final OpenBraille zeroth-review presentation dated 30-07-2026

22.1 Current official specification and price references

- Espressif, ESP32-WROOM-32 data sheet: https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32_datasheet_en.pdf
- Espressif, ESP32-S3 product page and DevKitC-1 guide: <https://www.espressif.com/en/products/socs/esp32-s3>
- TowerPro, SG90 analog servo specifications: <https://towerpro.com.tw/product/sg90-analog/>
- Indian price snapshots and procurement cautions are recorded in the Master Evidence, Costing and Decision Basis and must be refreshed on purchase day.

Prepared for academic project planning. The future research section is a management and experimental roadmap, not legal advice, patentability clearance, freedom-to-operate clearance, a filing recommendation or a promise of patent grant.